



Risk management and digitalisation to overcome barriers for safe reuse of urban wastewater for irrigation – A review based on European practice

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ABSTRACT

Despite their multiple benefits and several successful case studies, water-reuse practices are still not systematically implemented Europe-wide and globally mainly owing to the lack of social trust and high governance complexity in water-reuse techniques and practices. Non-homogeneity in the water-reuse regulations is also highly relevant. To overcome these challenges, the European Parliament has recently approved the Regulation 2020/741 on minimum requirements for water reuse to standardise the legal requirements of wastewater to be considered fit for reuse. This regulation also requires elaboration of Water Reuse Risk Management Plans (WRRMPs) for all reclamation facilities of the Union. This review aims to provide a general overview of the current limitations and gaps in systematically applying water-reuse practices, as well as recommendations that could overcome them. To this end, WRRMPs and digital solutions can play a crucial role because they can ensure the safe use of reclaimed water in agriculture and support communications to stakeholders, end-users and citizens. Guidelines to develop risk management plans and the current regulations regarding water reuse in Europe have also been extensively analysed in this study together with examples of digital support to risk management in water reuse.

1. Introduction

Water scarcity is one of the greatest challenges worldwide. Approximately 40 % of the global land area is classified as arid, semi-arid, or dry sub-humid, and over two billion people experience high water stress [1,2]. The situation is deteriorating because of climate change and increasing population. The agricultural sector has the largest water use, accounting for approximately 70 % of the total extracted water [3]. Therefore, alternative water sources must be explored to ensure adequate water supply. In this context, appropriately treated municipal wastewater (termed as reclaimed water) can help to compensate for the gap between available and required water worldwide, providing a regular water supply [4]. Wastewater reuse in agriculture involves numerous environmental and social benefits, both monetary and non-monetary. It reduces the exploitation of natural water resources, decreases pressure on their water basins, improves the ecological status of water bodies, alleviates poverty in water-scarce developing countries, and ensures socio-economic and political stability [5–8].

Approximately 312,000 hm³ of municipal wastewater is estimated to be produced worldwide annually [9]. Conventional municipal

wastewater treatment plants (WWTPs) have been designed with the aim to reduce the concentration of some pollutants (mainly organic matter, suspended solids and nutrients) below limits established in the legislation. Hence, they follow a linear approach where resources contained in the wastewater are not utilised. However, WWTPs can be substituted by (or transformed into) reclamation facilities to maximise the production of reclaimed water, combined with energy and nutrient recovery, and transition towards a circular economy [9,10]. Despite its substantial potential benefits, the percentage of wastewater reused (instead of only treated and discharged) remains low, except for certain cases, such as in Cyprus (Table 1).

The limited use of reclaimed water is often related to the risks associated with the pollutants contained in wastewater effluents. This is more significant in developing countries, where wastewater is partially treated or even untreated [3,5]. In addition, numerous end-users and the general public commonly consider wastewater as waste rather than a source for reclaiming water. To shift this paradigm, ensuring safety of the environment and public health in the use of reclaimed water is essential. These are the main goals of the Water Reuse Risk Management Plans (WRRMPs), which are mandatory for all reclaimed water facilities

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implemented in the European Union (EU) after applying the Regulation 2020/741 on minimum requirements for water reuse on 26 June 2023 [12]. This will be discussed in detail in Section 3. Apart from this, the implementation of digital technologies appears as key factor to improve current water management and ensure a safe reuse of treated water, as they can help to improve current water-reuse processes, management and policies [13,14].

This review aims to highlight the significance of risk management and digitalisation in the implementation of reclaimed water production, focusing on the European context. To this end, this paper is divided into the following sections: Section 1 is the introduction of the topic; Section 2 analyses the role of risk assessment in water reuse, including a review of the most relevant guidelines for risk assessment; Section 3 provides a critical discussion regarding the legislation, focusing on the innovative aspects and major gaps of the European Regulation 2020/741 [12]; Section 4 evaluates the main techno-economic and socio-political barriers that currently limit the widespread use of reclaimed water in Europe; and Section 5 describes the importance of digitalisation in boosting water-reuse practices, including the description of some innovative digital solutions.

2. Risk assessment in water reuse

Risk represents the likelihood of a hazard (biological, chemical, physical, or radiological), causing harm within a specified timeframe, including the severity of the consequences.

Risk assessment is the key step of management and implementation plans for water reuse and consists of the identifying and characterising possible hazardous events, estimating their likelihood, evaluating their impacts and their related risks and prioritising them [15,16]. The level

of detail and methodology used to assess risks may vary according to the specific needs and data available.

Risks associated with water reuse in irrigation can be divided into three categories: i) health risks associated with consumers (including animals) of agricultural products irrigated with reclaimed water; ii) health risks associated with humans exposed to reclaimed water (workers, bystanders, and residents in nearby communities); and iii) risks to the local environment, that is, surface waters, groundwater, soil, and ecosystems [5,7,17,18].

Minimising the health and environmental risks associated with water reuse is a complex task that requires a comprehensive knowledge of the processes, practices and value chains. Owing to the scarce specific regulations and standard procedures for risk management, the main international organisations involved in public health and environmental protection have provided guidelines and manuals to address risk management for water-reuse projects. However, this is approached from different perspectives based on the specific case at a specific time. Table 2 shows a summary of guidelines and manuals that have been selected due to their relevance in the water-reuse sector.

Some guidelines, such as the UNEP [19], do not specify a methodology for risk assessment, limiting the provision of maximum values and barriers to ensure safe water reuse. In contrast, most existing guidelines suggest sophisticated microbial analyses, epidemiological studies, and quantitative microbial risk assessment (QMRA) and/or chemical risk assessment (QCRA) approaches to be used when data and resources are available [27]. QMRA guidelines [24] provide information, data, and formulae to evaluate microbial risk and compare them with the target disability-adjusted life years (DALYs). Moreover, health risks due to chemicals (QCRA) can be assessed quantitatively by comparing estimated exposures to tolerable (or acceptable) effect levels or

Table 1

Worldwide data related to the production, collection, treatment and reuse of wastewater for agricultural purposes [11].

Area	Country	Produced municipal wastewater (hm ³ ·y ⁻¹)	Collected wastewater (%)	Treated wastewater (%)	Reuse of Treated Wastewater (hm ³ ·y ⁻¹)	Reused wastewater (%)	Reused wastewater in agriculture (%)	Water withdrawal for irrigation (hm ³ ·y ⁻¹)
Asia	China	48,510	64.2	100	7720	15.7	16.3	385,200 ^a
	Japan	16,930	71.00	68.3	390	3.4	3.0	53,300
Africa	Egypt	7078	91.8	60.5	2400	56.0	12.1	45,110
	Tunisia	312	88.8	87.8	42	15.5	33.0	1552
Central Europe	South Africa	2420	100	90.9	3220	–	–	11,397
	Ukraine	1513	–	59.1	6	0.7	–	4052
–Eastern Europe	Romania	1011	96.34	88.2	–	–	–	0,441
	Russian Federation	12,320	71.43	74.7	–	–	–	7000
Southern Europe	Croatia	336	100	83.7	–	–	–	30
	Poland	2168	96.36	61.3	6	0.5	–	84
Cyprus	Cyprus	35	100	96.5	34	99.7	50.0	174
	Italy	3926	–	99.4	90	2.3	96.7	16,000
Malta	Malta	26	100	92.3	1	5.8	50.0	24
	Portugal	577	100	46.8	6	2.2	–	3395
Spain	Spain	5206	100	90.0	1184	25.3	–	18,645
	Denmark	368	100	100	–	–	–	51
Western Europe	France	4000	94.25	94.25	822	21.8	–	2588
	Germany	5,213 ^b	100.0	99.42	–	–	–	0,363
UK	UK	4089	99.0	99.0	328	8.1	–	84
Middle East	Iran	3548	32.8	24.9	328	37.1	–	86,000 ^a
	Israel	0,500	100	90.0	1028	–	–	1,173 ^a
Saudi Arabia	Saudi Arabia	1546	74.0	100	508	31.8	100	19,000 ^a
Northern America	Mexico	7960	84.5	54.9	1796	41.1	76.8	65,870
	Canada	6074	97.3	95.5	–	–	–	2950
USA	USA	60,410	78.2	75.1	5548	12.2	–	163,000
	Argentina	2458	64.9	11.8	182	62.8	–	27,930 ^a
Southern America	Brazil	10,081	65.7	51.9	100	1.8	8.0	32,160
	Colombia	2397	100	6.4	–	–	–	16,000
Oceania	Australia	2094	87.3	95.5	280	14.0	50.0	11,099

^a Referred to annual quantity of self-supplied water withdrawn not only for irrigation, but also for livestock and aquaculture purposes.

^b Amount of wastewater collected.

Table 2

Summary of the information provided by guidelines for water reuse and safety plans.

	UNEP [19]	WHO Guidelines [16]	Australian Guidelines [15]	EPA [20]	ISO 16075:2020 [21]	ISO 20426:2018 [22]	SSP [23]	QMRA [24]	DEMOWARE [25]	JRC [26]
Water reuse practices	- Urban and agricultural reuse. - 4 quality classes.	Agricultural reuse of wastewater, greywater and excreta	- Urban reuse. - Irrigation in agriculture. - Industrial reuse	- Urban. - Industrial. - Agricultural. - Impoundment. - Environmental. - Groundwater recharge. - Potable reuse.	Agricultural and urban irrigation projects.	Non-potable reuse: agricultural, urban, industrial, recreational and environmental	Agricultural reuse of wastewater, greywater and excreta	Water Safety Management. all uses.	All possible reuses	Agricultural reuse of urban wastewater
Water quality criteria	- Intestinal nematodes. - <i>E.coli</i> . - Faecal coliforms. - Suspended solids.	- Reference pathogens: Campylobacter, Cryptosporidium, Rotavirus. - Risks expressed in DALYs.	Reference pathogens, soil and plant toxic compounds. Risk expressed in DALYs.	Faecal coliforms, BOD, Turbidity, pH, residual Cl_2 , metals and soil characteristics	BOD, TSS, Turbidity, thermotolerant coliforms, intestinal nematodes, organic matter, nutrients salts and heavy metals	<i>E.coli</i> , BOD, TSS or Turbidity Chlorine residual	Reference pathogens (e.g., Campylobacter, Cryptosporidium, Rotavirus). Risk expressed in DALYs.	Reference pathogens (e.g., Campylobacter, Cryptosporidium, Rotavirus). Risk expressed in DALYs.	Microbial, toxic, Compounds of agronomic relevance, and Compounds of emerging concern (no limits)	<i>E.coli</i> , BOD, TSS or Turbidity. Additional parameters as suggested by ISO and Australian guidelines [15,22].
Risks addressed	Microbial	- Health risks (priority) - Environmental	Health and environmental	Health and environmental	Health and environmental	Health	Health	Microbial	Health risks (priority) - Environmental aspects	Health risks (priority) - Environmental aspects
Risk assessment	No. Reuse based on limit values.	Microbial analysis, Epidemiological study, QMRA	Qualitative and Quantitative risk assessment	No. Reuse based on limit values. Methods for assessment specified in other documents [88]	- No. Reuse based on limit values	Qualitative and Quantitative health risk assessment	Qualitative and Quantitative risk assessment.	Quantitative risk assessment	Qualitative and Quantitative risk assessment	Qualitative and Quantitative risk assessment
Risk management	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach	Multi-barrier approach
Monitoring	Minimum frequencies and sampling points	Validation, Operational, Verification	Validation, Operational, Verification at critical control points	Frequencies depending on intended use.	Operational monitoring on wastewater, crops, soil and environment in ISO 16075 [21]	Parameters to be monitored and log removals	Validation, Operational, Verification	–	Operational	Operational monitoring, environmental monitoring
Social aspects	Not contemplated	- Public perception. - Economic feasibility. - Policy aspects	- Acceptance. - Communication strategies.	- Public Outreach. - Participation, and Consultation.	Not contemplated	No contemplated	Supporting programs	No contemplated	- Acceptance. - Communication.	Coordination and communication
Advantages	Specific for the most common practices in the Mediterranean region.	Referenced data, e. g., epidemiologic studies, concentrations in wastewater, crops and soil, log removals.	Complete of referenced characterisations, limits, examples and recommendations for risk management	- Referenced data - Provide suggestions from the planning and management phase to the public participation and consultation	Provide design and technical specifications for treatment, storage, distribution and irrigation infrastructure.	Provide guidelines to evaluate qualitative and quantitative risk assessment	Provide examples	Address the core phase of Risk assessment in a quantitative approach	Simplify the structure of a water reuse safety plan	Technical guidelines developed in accordance with EU Regulation 741/2020

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Table 2 (continued)

	UNEP [19]	WHO Guidelines [16]	Australian Guidelines [15]	EPA [20]	ISO 16075:2020 [21]	ISO 20426:2018 [22]	SSP [23]	QMRA [24]	DEMOWARE [25]	JRC [26]
				phase. - Address the existing regulatory framework to minimise the related risk - Discuss the funding possibilities available for water reuse projects.	No social aspects included. No guidelines on risk assessment.	No social aspects included. No limit values specified	- Focused on developing countries.	Do not consider non-technical aspects, monitoring and implementation	- No quantitative indications.	May have to be implemented with specific (National) requirements
Disadvantages	- Not addressed all possible reuses. - Not addressed environmental risks.	Focused mainly on developing countries.	- Complex. - Lot of data required	- Focused on US applications.	No social aspects included. No guidelines on risk assessment.	No social aspects included. No limit values specified	- Focused on developing countries.	Do not consider non-technical aspects, monitoring and implementation	- No quantitative indications.	May have to be implemented with specific (National) requirements

DALYs: Disability-adjusted life years; EPA: United States Environmental Protection Agency; JRC: Joint Research Centre-European Commission; QMRA: quantitative microbial risk assessment; SSP: Sanitation Safety Planning; UNEP: United Nations Environment Programme; WHO: World Health Organisation.

concentrations [25]. In this respect, risk quotients (RQ) represent the ratio between the concentration at the endpoint and the respective tolerable concentration or dose. An $RQ > 1$ indicates that, based on current knowledge, risk is above the acceptable levels, hence reduction measures should be implemented. However, an $RQ < 0.1$ is often defined as areas of negligible risk [28]. Even if the quantitative assessment follows the same approach both for microbial and for chemical risks, the implementation of a QCRA may be more complex. Guidelines do not cover in detail the procedures and the evaluation of target concentrations should consider toxicological outcomes, short-term and long-term effects and simultaneous exposures and combined effects [27]. At European level, the most recent and comprehensive guidance elaborating the 11 key elements of the risk management plans required in Regulation 2020/741 [12] has been published by the Joint Research Centre (JRC) in 2022 [26]. They aim to fill the gaps of the Regulation on the elaboration of WRRMPs (Section 3.2).

2.1. Integrated nutrient management in water reuse

In line with the current European policies (such as the new Circular Economy Action Plan) that aim to encourage circular approaches in the water sector by increasing safe wastewater reuse, the European Commission (EC) aims to develop, apart from the mandatory WRRMPs commented before, Integrated Nutrient Management Plans (INMPs) to stimulate nutrient recovery [29] (EC-European Commission, 2020). The combination of water reuse with the recovery of the nutrients contained in wastewater for agricultural purposes is commonly known as fertigation [7]. Fertigation could be of great interest as it enables not only to reduce the water needs of crops, but also to decrease (or even avoid) the use of synthetically produced nutrients that require approximately 10 kWh·kg-NH₄¹ and 20.7 kWh·kg-PO₄¹ [30]. In this context, Jiménez-Benítez et al. [6] reported positive economic and environmental balances when a scenario based on fertigation of anaerobic membrane bioreactor effluents was considered, i.e., a return of up to 376 k€·y⁻¹ and savings in CO₂ emissions of up to 898.9 tCO₂·y⁻¹ for mid-sized WWTPs. Similarly, Foglia et al. [31] evaluated water-reuse and fertigation scenarios in the Peschiera Borromeo WWTP (Italy), obtaining lower impacts on climate change (-28 %), fossil fuel depletion (-31 %), mineral resource depletion (-52 %), and freshwater ecotoxicity (-35 %) compared to a baseline scenario using a conventional activated sludge system. But excessive addition of nutrients can lead to environmental and health issues, such as crop-related problems, nutrient leaching, and runoff, which can result in groundwater pollution and/or eutrophication of water bodies [8,32]. The goal of INMPs is to ensure good fertigation practices by properly regulating nutrient addition to soil, thus ensuring their safe use and avoiding health and environmental issues. According to Seco et al. [33], INMPs should contain the following elements: i) a water balance considering the effluents from reclaimed facilities, returns to water bodies through distribution pipelines, irrigation systems and fields, evapotranspiration, and crop requirements; ii) a nutrient balance that considers their concentrations in reclaimed water, nutrient losses associated with returns, variations in concentration due to evapotranspiration, crop nutrient requirements, and mineral fertilisation; and iii) an economic balance that accounts for different monetary flows within the reuse system, such as costs of purification, pumping, fertilisers, irrigation water, and management plans.

3. Water-reuse regulations

Some EU member states from the south (Cyprus, France, Greece, Italy, Spain, and Portugal) have established specific legislation to regulate water reuse in centralised WWTPs (Table 3). On the contrary, other members such as Austria, Germany, Denmark, the Czech Republic, the Netherlands, and Belgium have not regularised water-reuse practices yet [34]. Three main approaches regarding water-reuse regulations have been identified [17].

Table 3
Summary of relevant information of National Regulations on Water Reuse in some EU countries.

Country	Cyprus	France	Greece	Italy	Spain	Portugal	EU 2020/741
Standards reference	[39]	[40]	[41]	[42]	[43]	[44]	[12]
Issuing institution	Ministry of Agriculture, Natural resources and Environment Water development Department (Wastewater and reuse Division)	Ministry of Public Health Ministry of Agriculture, Food and Fisheries Ministry of Ecology, Energy and Sustainability	Ministry of Environment Energy and Climate Change	Ministry of Environment Ministry of Agriculture, Ministry of Public Health	Ministry of Environment Ministry of Agriculture, Food and Fisheries, Ministry of Health	Portuguese Environment Agency	European Parliament
Intended use	Urban areas irrigation, agriculture irrigation, golf course irrigation, aquifer recharge. Only from urban wastewater.	Urban areas irrigation, agriculture, golf course irrigation, green areas not accessible to the public	Urban areas irrigation and street cleaning, agriculture, industrial reuse, aquifer recharge, golf course irrigation, green areas not accessible to the public	Urban areas irrigation and street cleaning, agriculture, industrial reuse, golf course irrigation, green areas not accessible to the public	Urban areas irrigation and street cleaning, private gardens, agriculture, industrial reuse, aquifer recharge, golf course irrigation, green areas not accessible to the public, silviculture, environmental uses (e.g., wetlands, minimum stream flows)	Agriculture irrigation, urban uses (landscape, flushing, fire-fighting, street cleaning, recreational uses) or ecosystem support	Agriculture irrigation
Quality categories	5 Protection of waters against pollution caused by nitrates from agricultural sources	4	3	1 (apart from industrial reuse)	12	5 Technical Guide on wastewater reuse, to support the implementation of water reuse projects	4 Technical guidelines for water reuse schemes, to support the implementation of risk management plans
Regulated parameters (microbial)	20 (2)	6 (4)	up to 80*	55 (2)	Up to 90* (3)	21 (2)	4 (1 – Legionella and Helminths eggs only in specific cases)
Monitoring frequency	1/15 days; pH: 3/week; metals: 2/year; Chlorides: 1/month; Helminth eggs: 4/year.	1/week 1/two weeks 1/month	Turbidity, <i>E. coli</i> , total coliforms: 7-1/week ; others: 24/year 12/year 4/year	Regional regulations	Microbial: 3/week - 1/month; TSS: 1/day - 1/week; others: 1/two weeks - 1/month	1/week – 1/2 weeks, continuous operational monitoring	1/week 1/two weeks Continuous
Risk management	–	Risk analysis of treatment and distribution malfunctions	–	–	Guidelines on permitting procedures and technical support for risk assessment for health and environment	Guideline for risk assessment	Technical guidelines for risk management
Public	–	Information boards in green public spaces	–	–	–	Communication to public	Communication to public
<i>E. coli</i> (n/100 mL)	5-10 ³	250-10 ⁵	5- 200	10	0-10 ⁴	10-10 ⁴	≤ 10 - 1000
BOD5 (mg/L)	10-70	COD: 60	10-25	20	–	10-40	≤ 10 - 25
TSS (mg/L)	10-30	15	2-35	10	5-35	10-60	≤ 10 - 35
Turbidity (NTU)	–	–	2-no limit	–	1-15	≤ 5 – no limit	≤ 5 – no limit
Others	Helminth eggs, COD, pH, conductivity, Heavy metals and metalloids, Fat & oil, chlorides, nutrients	Faecal enterococci, Sulphate-reducing Bacteria, F-specific Bacteriophages	Total coliforms, pH, conductivity, Heavy metals and metalloids, chlorides, nutrients, TDS, SAR, bicarbonate, Toxic substances including priority substances	Salmonella sp., pH, conductivity, Heavy metals and metalloids, Fat & oil, chlorides, nutrients, SAR, Toxic substances including priority substances	Metals, EC, SAR.	Helminth eggs, Ammonia, Nitrogen, Phosphorus, Heavy metals, SAR, Salinity and Boron	Total coliphages (F-specific coliphages or somatic coliphages), <i>Clostridium perfringens</i> spores (or spore-forming sulphate reducing bacteria), Legionella spp.: <1000 CFU/L; Intestinal nematodes: ≤ 1 egg/L

*: Depending on the final use and specific project requirements.

**Lowest limiting value for Class A; highest limiting value for Class D.

- i) Based on limiting values that are defined for several parameters related to reclaimed water. Health and/or environmental risks can be minimised by meeting these values, which may be applicable at different points based on the standards (e.g. at the reclaimed water delivery point or facility outlet). This approach is followed by EU members such as Italy, Spain, France, Cyprus, Greece and Portugal [35] and requires intensive monitoring of water quality, which can be highly challenging when analysing emerging pollutants [36,37].
- ii) Based on wastewater treatment requirements and limiting values. This approach has been reported in Title 22 of California's Code of Regulations and followed by the EPA guidelines [20]. For each potential use, a specific approved and certified wastewater treatment technique is required. For certain use categories, the water quality criteria (limiting values) can also be applied.
- iii) Based on implementing a risk management system for each reuse project. This approach was adopted by the Australian guidelines [15] and WHO guidelines [16], followed by the DEMOWARE manual [25] and the more recent EU Regulation 2020/741 [12]. An advantage of this approach is the prior identification and management of risks more proactively, rather than relying on post-treatment analysis and response when problems have already arisen. It is also more flexible because it can be applied to various scenarios. The main health and environmental hazardous events must first be identified and assessed, and measures to prevent and control the risks must be implemented, followed by the implementation of monitoring procedures to ensure that risks are effectively reduced to an acceptably low level.

Amongst water-reuse regulations, apart from the approach to be followed, large disparities occur in the number of parameters to be analysed and their limiting values, as well as the sampling points and sampling frequency (Table 3). This situation is often related to the fact that the regulatory framework has not addressed water reuse holistically (considering all factors that may affect water reuse such as water, energy, food, climate, health, etc.), but it has been commonly managed separately [38].

To solve the conflicts related to the lack of standardisation in national legislations of EU members, and consequently, to increase the amount of reclaimed water used in the continent, the European Parliament and the Council recently approved Regulation 2020/741 on minimum requirements for water reuse [12]. This Regulation establishes quality limits for certain pollutants as well as the frequency of monitoring and treatment performance targets for reclaimed water reuse in agriculture, depending on four different quality classes in a fit-for-purpose approach. Water quality classes (Classes A–D) are defined in terms of pathogens, biochemical oxygen demand (BOD), total suspended solids (TSS) concentrations, and turbidity. Microbiological pollution is assessed by the *E. coli* concentration in water (bacterial indicator), combined with the total coliphages (alternatively, F-specific coliphages or somatic coliphages) and *Clostridium perfringens* spores (or spore-forming sulphate-reducing bacteria), which act as virus and protozoa indicators, respectively. The limiting values of these parameters varied according to the quality class (Table 3). The limits of *Legionella* spp. and intestinal nematodes are also indicated for certain reuse practices. Water-quality classification also depends on the crop category to be irrigated and irrigation technique. Class D can only be applied to non-food crops, Class C applies to food crops with above-ground edible parts and irrigation using drip irrigation, and Class B can use other irrigation methods. The most stringent is Class A which includes the irrigation of food crops consumed raw, which are in direct contact with reclaimed water [12]. Monitoring requirements also depend on the reclaimed water-quality class, in addition to the specific parameter analysed. *E. coli* monitoring varies from twice a month (Classes C–D) to once a week (Classes A–B), whereas BOD and TSS should be analysed once a week, and **turbidity** must be continuously monitored. As

monitoring is crucial to ensure that the water-reuse scheme effectively achieves defined performance and water quality targets, validation and routine monitoring should be performed before a new reclamation facility is put into operation.

The new EU Regulation also introduces WRRMPs as a mandatory step for determining the minimum requirements for specific pollutants in specific settings to reduce the risks associated with public health and the environment to the maximum extent. The WRRMP should identify and manage risks proactively to ensure that reclaimed water is safely used and managed, and poses no significant risks to the environment, human or animal health. The Water Reuse Risk Management Plan is based on the key risk management principles as defined in Annex II of Regulation 2020/741 [12]. Risk assessment can be performed using qualitative or semi-quantitative risk assessments. Quantitative risk assessment should be used when sufficient supporting data is available or in projects that pose a potentially high risk to the environment or public health. Based on the risk assessment outcomes, additional requests regarding heavy metals, contaminants of emerging concern (CECs), and/or antibiotic resistance genes/bacteria (ARG/ARB) are required.

3.1. Innovative aspects of EU Regulation 2020/741

EU Regulation 2020/741 fills a legislative gap by defining and standardising the requirements and conditions for water-reuse applications. One of the most innovative aspects of this regulation is the shift to the fit-for-purpose approach, instead of the fit-for-all method that has been addressed by some EU members such as Italy and Spain. Accordingly, not only are quality limits established, but crop quality and irrigation techniques must also comply. This fit-for-purpose approach aims to ensure health and environmental protection, while trying to facilitate the techno-economic feasibility of water-reuse applications [13]. This is supported by the fact that, compared to national legislation, the EU Regulation selected a lower number of parameters to be monitored to accomplish quality requirements (Table 3). However, it also provides EU members the flexibility to include in their national laws limiting concentrations of other pollutants that the members considered necessary to evaluate, such as ARB, antibiotics, microplastics, etc. These compounds have been scarcely considered in previous legislation but are specifically considered in this EU Regulation and the updated Directive concerning urban wastewater treatment (under revision) [45] because of the growing concern they generate.

Other innovative aspects of this regulation are as follows: i) monitoring requirements are explicitly expressed and therefore normalised, whereas in some national regulations, such as in Italy, they are provided by local authorities; ii) reclamation facility operators are defined, shifting the conventional paradigm of wastewater treatment to wastewater reuse; iii) the relevance of public awareness regarding the benefits of water reuse has been highlighted; iv) the promotion of fertigation; and v) the proposal of financial incentives for water reuse and further research into the topic [13].

3.2. Gaps and limitations in the regulation

Although Regulation 2020/741 has harmonised water-reuse practices providing standards and requirements, some aspects have not yet been thoroughly addressed. For risk management, key elements are listed but not described. The regulation also states that the methodology for risk assessment should at least be semi-quantitative, but no further information is provided. To overcome this, technical guidance has been recently launched by the JRC of the European Commission [26]. The structure is divided into 11 key risk management (KRM) elements grouped into four modules: I) preparation (KRMs 1–2); II) risk assessment (KRMs 3–6); III) monitoring (KRMs 6 and 9); and IV) governance, management, and communication (KRMs 7–11). Suggestions are provided, as well as supporting information and examples of good practices

of full-scale water reuse case studies. However, the specificity of each system may require ad-hoc considerations that cannot be generalised into a common scheme. **The general structure provided must be adapted case by case to the specific needs and the required level of detail may be much deeper and imply further studies and analyses.**

Some examples of control measures are provided in Regulation 741/2020 [12], but a method to evaluate their contribution to risk minimisation is not well defined. Moreover, the legislation specifies that the reuse system description must consider all the integrated systems from the wastewater source to its final use, including possible external contributions such as dilution of treated wastewater with other sources, but the regulation does not specify a technique to address these aspects. This creates a significant gap in some areas, such as Italy, where many distribution systems are managed by irrigation consortia that generally use surface water to irrigate crops whose quality is not monitored or regulated; thus, they do not have the infrastructure to undertake these tasks. All these uncertainties present the risk to create some controversies between the updated national water-reuse legislation of EU members when the Regulation 741/2020 is to be applied (26 June 2023).

Regarding microbiological risks, a detailed description of the method to assess them is provided. Nonetheless, indicators, such as *E. coli*, *Legionella* spp., and intestinal nematodes, are not always monitored in WWTPs. In many cases, they are only analysed in the effluent, being unable to know the total log removal of pathogens in the WWTP. Thus, it can be hard to ensure, due to the lack of data regarding the occurrence, trends, and treatment efficiencies in pathogen removal, that the treatment system could accomplish with the legislation (in terms of total pathogen removal), which may discourage the construction of new water reclamation facilities. Moreover, pathogen determination is expensive and can be affected by many variables, outer contamination of samples, and the inaccuracy of the equipment and measuring methodologies. In fact, standard analytical protocols include sampling, transportation, dilution, handling and storage practices, which require long response times (18–72 h) and may introduce errors [46]. Consequently, they sometimes provide results with high uncertainty and could imply an extremely long response time when pathogen proliferation occurs. Furthermore, some techniques for pathogen analysis are simply based on the presence of pathogenic microorganisms in the sample, and do not distinguish whether they are active or inactive. In cases where inactive biological pollutants are present, a reduction in their concentration is not directly related to a decrease in risk. All these limitations in current detection techniques hinder in attaining the legal limits and increase operating costs, primarily due to the requirement of higher doses of disinfecting agents [13], without necessarily improving safety in the reuse of water.

More innovative monitoring technologies and techniques that are supported by reliable digital solutions and tools must be developed (Section 5). However, the regulation only considers standard lab measurements to verify the accomplishment of quality requirements, limiting the use of digital tools to internal monitoring procedures.

In addition to the aforementioned limitations, information is lacking on a method to manage risks from other pollutants, such as CECs, ARG, and metals [4]. Considering them is extremely challenging because of the technical and economic difficulties in their monitoring. Some authors state that reclaimed facilities, end-users and other stakeholders should share these analyses' costs [47]. But this is hard to put into practice as no information is provided regarding cost sharing in the regulation and the willingness-to-pay for reclaimed water of end-users is usually low in many European regions [35,48]. No references have been made to reducing pollutants at the source of origin in the Regulation 741/2020, even though it is widely known to be one of the most relevant factors influencing the final water quality [7,49]. However, this gap can be filled in EU-member national legislations as the proposal for the updated directive concerning wastewater treatment stresses this issue [45]. Another important aspect (not specified in the regulation) is the monitoring requirements of soils and/or crops. No information is

provided regarding the parameters that should be monitored, frequencies, or sampling points. This is crucial for assessing the environmental impacts of water-reuse practices and evaluating potential health risks, particularly for toxic compounds that may accumulate in crops and humans [7].

According to the proposed regulation [12], determining the need for additional requirements requires the operator to compare the outcomes of the risk assessment to acceptable levels of risk or water quality. The Regulation 2020/741 provides no further guidance, except for referring to other EU regulations. Optimising these regulations or other environmental and health criteria to minimise requirements for reclaimed water facilities while ensuring the safe reuse of water is a complex task, even for water risk-assessment experts. Some authors have stated their concerns that these difficulties, combined with the associated costs of reuse, might unintentionally lead to indirect reuse [50]. Hence, it is essential to reduce barriers that could ease the wide implementation of water reuse practices. Some of these barriers are explained in Section 4.

4. Limitations to the widespread application of wastewater-reuse practices

The development of wastewater reuse has diverse challenges. Generally, governance complexity and limited modernisation of the water sector tend to hinder the transition to a more circular water management where wastewater reuse becomes central. Specifically, water reuse challenges include those related to the water reclamation process itself, that is, process feasibility and the presence of potentially damaging microorganisms and compounds in the reclaimed water, as well as socio-political limitations [13].

4.1. Limitations related to water-reuse processes

Reclaimed water presents intrinsic risks associated with the presence of biological and chemical pollutants in wastewater. Depending on the type of pollutant, their concentration, and their level of exposure, they can have negative impacts on public health and the environment. Consequently, some end-users feel reluctant to consume it. Risks cannot be eliminated completely but can be minimised to acceptable levels. The definition of acceptable could be partial, as it depends on socio-economic context, but with the goal to standardise it, the WHO [16,24] and Australian guidelines [15] has defined 10^{-6} DALYs as the health limit, which corresponds to one infection per one million people per year.

Current WWTPs are highly efficient in removing certain pollutants (organic matter, solids, nutrients) with the goal to accomplish with current legal limits that try to avoid serious environmental and health issues related to their emissions, e.g., oxygen depletion in water bodies, eutrophication, human diseases, etc. However, evidence on the mechanisms by which WWTPs effectively deal with other pollutants such as metals, CECs, and some pathogenic microorganisms such as viruses and ARG/ARB remain limited [51,52].

Pathogens cause human diseases, that can be both chronic and acute, and even premature mortality. They are thus of high health concern [5]. The presence of pathogens in wastewater effluent streams is extremely common, as many WWTPs lack highly efficient biomass-water separation systems. Consequently, referent pathogenic organisms to assess pathogenic risk in water reuse (such as *Escherichia coli*, *Campylobacter*, *Cryptosporidium*, Rotavirus, nematode eggs, and faecal coliforms) are included in the risk assessment guidelines (Table 2) and regulations related to water reuse (Table 3). To reduce risks of biological pollution, most of guidelines suggest the multiple-barrier approach to retain and remove them to a low-risk level. For this, refining systems (such as sand filters and membranes) and disinfection systems are commonly used as post-treatment in water facilities that aim to produce reclaimed water [13], showing significantly different pathogen removal efficiencies (Table 4).

Table 4
Microbial removal capacity of some wastewater post-treatment technologies.

Treatment	Technology	Removal efficiency [log removal]	Reference
Chemical	Chlorination	Bacteria: 1.5–4 Virus: 2–4 Protozoa: 2–3	[53]
	Peracetic Acid	Bacteria: 0.2–4	[54]
	Performic Acid	Bacteria: 1–2.7	[55,56]
	Ozone	Bacteria: 2–4 Virus: 2–4 Protozoa: 2–3	[53,57]
		Bacteria: 3–4.5	[58,59]
Chemical-Physical Physical	AOP	Bacteria: 0.4–4 Virus: 0–1.5 Protozoa: 0.2–3.5	[57,60,61]
	UV	Bacteria: 3.4 Virus: 4–5	[62]
	MF – UF – NF	Bacteria: 5–7	[62]
	AnMBR	Bacteria: 1.7–3.2	[62]
	AeMBR	Virus: 0.3–6.3 Bacteria: 1.8–2.7	[63]
Nature-based solutions (NBS)	Sandy subsurface flow	Bacteria: 1.1–3.2	[64,65]
	Constructed wetland	Bacteria: 0.2–1.9	[65]
	Lagoon		

AeMBR: aerobic membrane bioreactor; AnMBR: anaerobic membrane bioreactor; AOP: advanced oxidation process; MF: microfiltration; NF: nanofiltration; UF: ultrafiltration; UV: ultraviolet radiation.

Apart from post-treatment process, there are other barriers that aim to limit public exposure to reclaimed water such as retention time in storage systems, selected irrigation systems, setback distances from the application site, entry restrictions, avoiding or preventing cross-connections with potable water distribution systems or backflow prevention, and encoded distribution systems. All these measures aim to minimise the direct contact with reclaimed water that could present potential risks, especially for population with higher level of exposure, such as farmers, crop merchants, agricultural product consumers, and residents from surrounding communities. However, not many European regions currently have water management systems that can implement all those barriers efficiently as they often lack technical infrastructures and water policies to manage so. In developed countries (mainly Western and Northern EU members), additional infrastructure is often required for the transition from conventional WWTPs to reclamation facilities, which imply additional costs and difficulties that sometimes discourage the use of reclaimed water for irrigation [13,66]. For instance, based on the tertiary treatment (or post-treatment) technologies selected (Table 4), operating costs can increase by approximately 0.02–0.21 €·m⁻³ [67]. For comparison, the cost of extracting water from rivers and groundwater bodies is estimated to be only 0.015–0.2 €·m⁻³ [68]. In the case of less-developed EU members, the situation is often worse as wastewater collection is not always widespread and wastewater treatment systems are scarce or inefficient [5]. To reduce these economic barriers, a structural transformation of the water sector is urged, in a sense that the real value of water for our society could be recognised. This includes the recognition of the value of water in all our economic and societal activities (including the losses that could produce the lack of access to water), and of the resources contained in our water streams. Within this approach, the rational use and reuse of water is promoted [69].

Concerns are also increasing regarding the presence of CECs in water as they can affect human health and the environment (including flora and fauna) even at trace levels [18,52]. Moreover, they can accumulate in crop lands and the food chain [70]. Amongst all groups of CECs, antibiotics require special attention, not only because of their increasing presence in urban wastewater streams and the potential toxic effects of antibiotics entering the human food chain [71], but also because they

contribute to the development of ARG and ARB [36,51]. This has become a topic of major concern for the scientific community [72] because the increasing spread of ARGs threatens modern medicine and lifestyles, especially in developed countries that are highly antibiotic-dependent. However, knowledge related to long-term risks due to the presence of ARG or ARB in reclaimed water remains scarce [51,73], and most of these compounds are commonly stated as potential risks.

CEC removal in wastewater is central issue in the current proposal for the new Directive concerning urban wastewater treatment [45]. New limit values will be established for CECs that could require additional treatment and even the collection and treatment of wastewater in agglomerations smaller than 1.000 P.E. Those conditions would depend on the risk to the environment and human health in the surrounding areas. The extent of the risks associated to CECs depends on the degree of wastewater treatment (secondary, tertiary or quaternary), type of pollutant, and especially, on pollutant influent concentrations. Poorly treated industrial wastewater and/or runoff water in municipal wastewater streams can add significant amounts of pollutants, increasing the risks associated with its reuse [4]. Thus, pollutant reduction is crucial for improving the safe reuse of wastewater, especially from the source of production [49]. The risk assessment of CECs presents several challenges: i) detection and analyses of CECs are not always accurate because they are diverse and commonly present at trace levels [37] (Wilson and Aqeel Ashraf, 2018), ii) monitoring and analysis of CECs are complex and expensive due to the requirement of specialised equipment and the excessively high number of compounds to be analysed [36,74], iii) they are usually highly resistant to biological removal in WWTPs [52], iv) degradation compounds can present higher toxicity than the original CECs [4], and vi) they can be highly persistent and accumulate in the food chain [70]. To effectively assess the risks associated with CECs, guidelines that provide target compounds, detection techniques, and safe CEC concentration limits must be developed by experts, but they are currently limited.

4.2. Socio-political limitations

To be widely implemented, water-reuse practices must be accepted by facility operators and managers, authorities, users, consumers, and the civil population, who have to be aware of the water-reuse benefits and its possible risks and safety measures. This is highly relevant because a lack of trust can completely impede water-reuse development despite significant techno-economic advances [8,25,75]. Engaging stakeholders and the general public in planning of water-reuse systems is important (especially in at early stages), as it can help to align approaches; but this is not easy to take into practice. There exist some obstacles related to the water value chain governance, which is managed by high number and variety of public organisations. Although collaboration between these actors is intended, it is not always successful. Inadequate understanding of some roles and responsibilities often occurs, leading to conflicts and ineffective strategies [66]. Owing to these (and other) governance issues, a lack of homogeneity in national regulations regarding water reclamation has been observed (Table 3), which has been a significant barrier to the widespread development of water reuse in Europe [13,33]. Differences in regulations include the type and number of monitored pollutants, as well as their limiting values, which can be even as stringent as those for drinking water in certain regulations, such as in Italy. This lack of homogeneity also hinders transport and supply of agricultural products irrigated with reclaimed water [66].

As aforementioned, Regulation 2020/741 was launched with the goal to align policies and government decisions in the water sector, and the expectation to boost water reuse for irrigation in Europe [12]. However, both Regulation 2020/741 and the water sector still present some gaps and controversies that make important part of the population be reluctant to trust in water reuse [8,75]. Increasing the awareness regarding the benefits of water reuse, combined with improving transparency and ensuring the safety and low impacts of water reuse, are thus essential to

shift this paradigm and expand water reuse practices worldwide [13]. Risk assessment and management plans are clearly aligned with these goals. As they aim to ensure safe water reuse and provide the population with (in-time) data, the reputation and perception of reclaimed water facilities can be maintained at high levels [76], helping to change the vision of water reuse as risky practice.

Other social factors such as religious beliefs, unawareness of water scarcity issues or negative opinions about water reuse are also worth being considered because they can threaten the technical, economic, political, and legal advances that can be made in the sector [48]. For instance, in a case study analysed in Tunisia, political and economic efforts were made to reduce the costs of acquiring reclaimed water to 0.018 €·m⁻³ (lower price than surface water, i.e., 0.026–0.092 €·m⁻³). However, the farmers were reluctant to use it and preferred to use conventional water [77]. In Australia, owing to a serious water scarcity period during 2007–2009, a substantial investment (ranging billions of dollars) was made to build and implement reclaimed water facilities. However, after heavy rain episodes, concerns about water scarcity in the country vanished and these facilities got underused. Similarly, in France, water reuse is restricted to certain regions because this country only suffers from local and seasonal episodes of water scarcity [32], whereas Northern European countries hardly adopt water-reuse practices. Countries should not wait for serious scarcity events to develop political strategies based on safe water reuse. Building new water reclamation facilities, as well as the training and education of operators, technicians, managers and farmers, can take several years. Immediate action is necessary to be prepared for water scarcity episodes that will occur in the next decades and ensure safe water availability to all users, which is one of the sustainable development goals (SDGs) of the United Nations.

Table 5 summarises some of the aforementioned limitations that nowadays hinder the implementation of water reuse practices in the EU, as well as some suggestions to overcome them.

5. The role of digitalisation in water reuse

Digitalisation is acknowledged as key enabling factor to improve current water management and ensure safe reuse of treated wastewater, maximising its potential as resource. Digital technologies, which are nowadays more and more available, can help to develop control systems and produce near real-time monitoring data, combined with tools to treat, process, and analyse the data [14]. Hence, they have the potential to improve factors such as the efficiency and resilience of water infrastructures, as well as to extend their life significantly; encourage decarbonisation and circularity; optimise the workforce; reduce the

number of mistakes; rapidly detect malfunctions and anomalies during operation; improve the quality of water services; increase the speed in decision-making and raise the levels of awareness and collaboration between facilities, authorities and citizens [13,79]. Some innovative digital solutions implemented (and even demonstrated) in the wastewater sector are provided in Table 6.

However, the transition to digital water-smart wastewater sector is restrained by different aspects, such as: i) the fragmentation and lack of standardisation of the water market and policies; ii) the relatively small size of the water market; iii) issues related to the lack of trustworthiness in data sharing and cybersecurity; iv) high investment and/or technical difficulties to acquire new digital tools and fully integrate them into current functioning systems; v) high efforts (in terms of dedication and costs) to keep digital systems updated; vi) lack of integration of water security into the EU policy framework; vii) low social awareness on digitalisation; viii) scarce tangible evidences of some of the benefits provided by digital solutions; ix) low consumer demand; x) scarce alignment between advances in digital solutions and regulatory frameworks [13,26,79].

With the goal to overcome these barriers, the European Research Executive Agency (REA) has reported a series of key recommendations [26]: i) to boost the use of digital tools to prevent microbial pollution; ii) to encourage the creation of a volume market focusing on areas where the greatest demand is expected; iii) to align the procedures of relevant standardisation entities with the high speed of the market; iv) to support water operators in the digital transition; v) to define common data models and generate open data in and for public data services, also enhancing trustworthiness in data sharing; vi) to involve stakeholders in a transparent process of implementation analysis of water policies; vii) to raise awareness and promote training activities at all levels on digital solutions and cybersecurity; viii) to create programmes on advanced data management at local/regional scale; ix) to develop EU regulations for monitoring pollutants from stormwater; x) to strengthen the alignment between SDGs and the water framework directive; xi) to promote and standardise complex concepts such as Water-Energy-Food-Ecosystem (WEFE) Nexus with the aim to comprehensively assess the sustainability of water reuse; and xii) to promote research [26,79].

6. Conclusions

The reuse of wastewater for irrigation or fertigation is currently far from its potential use. Lack of social trust, non-homogeneity in the water-reuse regulations and high governance complexity appear as relevant factors hindering its development. The EU Regulation 2020/

Table 5

Description of current limitations to the wide implementation of water reuse (in Europe) and suggestions to overcome them.

Aspects	Limitations	Suggestions to overcome the limit	References
Governance/Legislative	Inadequate understanding between actors involved. Lack of standardisation. Normative is not always as modernised as the progresses in water sector. Requirements are not always detailed for each specific case.	Governance's incentives to water reuse practices, with (e.g., discounts on water tariff, green public procurement). Penalisation in the use of non-circular water sources. Clear and standard regulatory framework and assessment tools.	[33,66,78]
Process feasibility and performance	Additional infrastructure is often required to implement reclamation facilities, which imply additional costs. More complexity in the processes. Training of personnel. Lower efficiency and lack of implementation of systems for efficient removal of CECs and/or resistant pathogens. Limited spread of digital tools.	Digital tools to support process control and risk management. Urgence in implementing water recovery technologies to adapt to climate change and water scarcity. Direct or indirect support from authorities (e.g., subsidies, green public procurement, loans, etc.).	[13,51,52,69]
Social acceptability	Water reclamation practices may not be accepted by users or consumers. Lack of knowledge on the benefits of water reuse to mitigate water scarcity. Unawareness of water scarcity issues.	Improvement of transparency and assurance of safety and low impacts in water reuse. Plan communication and information to increase public awareness and boost policy towards sustainable practices.	[8,13,75]

Table 6

Examples of digital tools applied to water reuse.

Digital tool	Description	Integration into risk management	Benefits	Limitations	Reference
Improved network of sensors along treatment layout	Implementation of conventional probes (e.g., pH, ammonia, nitrates, TSS) with innovative sensors (e.g., TOC, UV absorbance)	Monitoring	Treatment optimisation and energy savings by facilitating rapid responses through real-time monitoring.	Sensor maintenance. Technical skills for big data analysis.	[14,80,81]
Sensor for Bacteria determination	Sensor for Bacteria monitoring	Monitoring	To assure the absence of bacteria or detect their presence in real-time.	Sensor outliers. Need calibration for wastewater matrix. Verification with standard analytical methods is needed for legal authorisations and for sensor calibration.	[46]
Early Warning System (EWS)	ANN and soft-sensing techniques for water quality monitoring and prediction	Control measure for monitoring and decision support	To provide real-time signals and facilitate rapid interventions when possible hazards are detected. They can rapidly detect system malfunctions or anomalous event occurrences, thus overcoming technical barriers related to delay in data acquisition from grab samples due to the lag time between sampling, measuring, and data analysis. It enables to use affordable sensors to predict trends in critical parameters that are not usually monitored or for which faulty signals may be generated. EWSs can be easily integrated into existing monitoring networks of WWTPs, without overloading analytical laboratories or installing expensive sensors. It can be used to support decision making.	Need to be validated according to the installed sensors. Prediction reliability is strictly depending on the availability and affordability of sensors' data.	[80–82]
WebGIS platform	Integrated platform	Monitoring	Improved management for water utility. Allows a complete overview of risks.	Require technical skills and periodic data updates, information must be elaborated	[83]
Match making tool	Web application containing information about reclaimed water quality, water availability, crop needs and water demand	Communication between water utility and users	Share information to match between water demand for irrigation and safe water availability	Distribution network for irrigation must physically connect WWTP with users. Requires data from agriculture practices and irrigation methods from different potential users.	[83]
Serious Game	Web application to simulate different water reuse scenarios and see the results in terms of water reuse, carbon emissions, energy consumptions and agricultural practices	Communication to public	To show the Nexus between water reuse, agriculture, carbon and energy footprint. To involve and train stakeholders and the general public in more sustainable water management approaches. To help decision-makers analyse different political options for the future.	It requires local and territorial data to develop realistic scenarios.	[38,84,85]
Tools for assessment methodologies (LCA, Nexus modelling, etc.).	Tools to develop standard assessment methodologies that can help to quantify the benefits of water reuse practices and analyse their risks.	Decision support. Modelling of water-reuse scenarios. Communication to public.	To quantitatively demonstrate benefits of water reuse compared to conventional treatment schemes. Raising the awareness of decision-makers, stakeholders, and the general public.		[6,30,31,38]
Tools for integrated sewer and wastewater system	Urban Wastewater System (UWS) modelling consider the interconnections amongst catchment, sewer network, WWTP and river system following a holistic approach.	Different sub-models are simulated and interconnections are verified through mass-balances; Performances of each sector are assessed using evaluation indexes.	Implement water quality-based objectives, uncertainty and sensitivity analysis in simulation scenarios of WWTP processes.	Model complexity, tools usually not open source.	[86,87]

ANN: artificial neural network; GIS: geographic information system; LCA: life cycle assessment; TOC: total organic carbon; TSS: total suspended solids; UV: ultraviolet; WWTP: wastewater treatment plant.

741 was recently approved to overcome some of these challenges and standardise the legal requirements of reclaimed water to be considered fit for reuse. A novel aspect of this regulation is the need to elaborate on WRRMPs in all reclamation facilities to minimise health risks and protect the environment when reclaimed water is used. This is also expected to transform people's opinions and increasing their awareness of the benefits of reusing water while reducing their biased opinions on using reclaimed water. Management plans will be also integrated with nutrient management plans in the near future to ensure safety in fertigation practices. However, the Regulation 2020/741 still presents some gaps such as the lack of information on how to implement standard WRRMPs, which could create some future controversies when the Regulation is to be applied. Hence, technical guidance has been recently launched by the JRC to standardise the elaboration of WRRMPs.

Apart from risk assessment, digital solutions are recognised as key enabling factors to improve water-reuse processes and management as they can help to develop control systems as well as to monitor and analyse the data. However, the transition to digital water-smart wastewater sector is not well-established yet and some efforts have to be done to overcome current barriers to its implementation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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